MATHGENIUS

ALGEBRA FUNDAMENTALS

A University-Level Study Guide

Definitions • Explanations • Formulas • Worked Examples • Practice Questions

Solve Smarter. Learn Better.

How to Use This Guide

Study one section at a time. Read the explanation, work through each example with pen and paper, then attempt the practice assessment before checking the answer key. This guide supports independent learning and revision; follow your institution's course outline and lecturer guidance.

Learning outcomes

- Use algebraic notation and translate words into expressions.
- Simplify, expand and factorise algebraic expressions.
- Solve linear, simultaneous and quadratic equations.
- Apply laws of indices, surds and polynomial operations.
- Solve inequalities and work with functions and graphs.
- Model practical situations using algebra.

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1. Algebraic Language and Number Systems

Variables and constants

A variable is a symbol representing a value that may change or is unknown. A constant is fixed. In $7x + 4$, x is the variable, 7 is its coefficient and 4 is the constant term.

Terms and expressions

Terms are separated by plus or minus signs. Like terms have identical variable parts and powers: $3x$ and $-8x$ are like terms, but $3x$ and $3x^2$ are not. An expression has no equality sign; an equation states that two expressions are equal.

Number systems

Natural numbers are counting numbers (whether zero is included depends on convention). Integers include negative whole numbers and zero. Rational numbers can be written as a ratio of integers. Irrational numbers, such as $\sqrt{2}$ and π, cannot. All are real numbers.

Translating words

Five more than x is $x + 5$. Twice a number decreased by 3 is $2x - 3$. Four less than a number is $x - 4$; 'less than' reverses the order.

Example: In $5a - 2b + 9$, the terms are $5a$, $-2b$ and 9. The coefficients are 5 and -2; 9 is the constant.

2. Operations and Algebraic Laws

Order of operations

Evaluate brackets first, then powers and roots, then multiplication and division from left to right, and finally addition and subtraction from left to right.

Commutative and associative laws

Addition and multiplication are commutative: $a + b = b + a$ and $ab = ba$. They are also associative: $(a + b) + c = a + (b + c)$, and $(ab)c = a(bc)$. Subtraction and division are not generally commutative or associative.

Distributive law

Multiplication distributes over addition: $a(b + c) = ab + ac$. Apply the multiplier to every term inside the brackets.

Fractions

To add or subtract fractions, use a common denominator. To multiply, multiply numerators and denominators. To divide by a fraction, multiply by its reciprocal.

Worked example: Evaluate $18 - 2(3 + 4)^2 \div 7$. First $3 + 4 = 7$; then $7^2 = 49$; then $2 \times 49 \div 7 = 14$; finally $18 - 14 = 4$.

3. Expressions and Simplification

Collecting like terms

Combine only terms with the same variable part and powers. For example, $7x + 4 - 3x + 9 = 4x + 13$.

Substitution

Replace each variable with its stated value, including negative signs in brackets, then follow the order of operations.

Algebraic fractions

Find a common denominator before adding or subtracting. For example, $2x/5 + 3x/10 = 4x/10 + 3x/10 = 7x/10$.

Restrictions

An expression with a variable in its denominator is undefined wherever that denominator equals zero. Record those excluded values.

Worked example: Evaluate $2a^2 - 3b$ when $a = -2$ and $b = 5$: $2(-2)^2 - 3(5) = 2(4) - 15 = -7$.

4. Expansion and Factorisation

Expansion

Use the distributive law to remove brackets. For example, $3(2x - 5) = 6x - 15$.

Two brackets

Multiply every term in one bracket by every term in the other: $(x + 4)(x + 2) = x^2 + 2x + 4x + 8 = x^2 + 6x + 8$.

Common factor

Factorisation extracts a common factor. For example, $6x^2 + 9x = 3x(2x + 3)$.

Difference of squares

$a^2 - b^2 = (a - b)(a + b)$. For example, $x^2 - 25 = (x - 5)(x + 5)$.

Perfect-square identities

$(a + b)^2 = a^2 + 2ab + b^2$; $(a - b)^2 = a^2 - 2ab + b^2$.

Worked example: Factorise $2x^2 + 7x + 3$. Find two numbers with product $2 \times 3 = 6$ and sum 7: 6 and 1. Split the middle term: $2x^2 + 6x + x + 3 = 2x(x + 3) + 1(x + 3) = (2x + 1)(x + 3)$.

5. Linear Equations and Applications

Balancing equations

A linear equation in one variable has the variable to the first power. Perform the same operation on both sides to isolate the variable.

Equations with brackets

Expand brackets, collect variable terms on one side and constants on the other, then divide by the coefficient.

Equations with fractions

Multiply every term by the lowest common denominator to remove fractions, while respecting any excluded values.

Word problems

Define the unknown, translate the information into an equation, solve, check the result and state the answer with units.

Worked example: Solve $2(3x - 1) = 4x + 10$. Expand: $6x - 2 = 4x + 10$. Subtract $4x$: $2x - 2 = 10$. Add 2 and divide by 2: $x = 6$. Check: $2(18 - 1) = 34$ and $24 + 10 = 34$.

6. Simultaneous Linear Equations

Meaning

Simultaneous equations must be satisfied by the same values of the unknowns. Their solution is the intersection of the corresponding graphs.

Elimination

Multiply one or both equations if needed so that one variable has opposite coefficients. Add the equations to eliminate that variable, solve, then substitute back.

Substitution

Rearrange one equation to make one variable the subject. Substitute that expression into the other equation, solve, then find the remaining variable.

Number of solutions

Two intersecting lines have one solution; distinct parallel lines have none; identical lines have infinitely many.

Worked example: Solve x + y = 11 and x - y = 3. Add: 2x = 14, so x = 7. Substitute: 7 + y = 11, so y = 4.

7. Indices, Surds and Scientific Notation

Index laws

For suitable non-zero a: a■ × a■ = a■■■; a■ ÷ a■ = a■■■; (a■)■ = a■■; a■ = 1; a■■ = 1/a■.

Fractional indices

$a^{1/2} = \sqrt{a}$ and $a^{1/3} = \sqrt[■]{a}$. More generally, $a^{■/■}$ represents an nth root raised to the mth power, subject to real-number domain restrictions.

Surds

A surd is an irrational root kept in exact form. Simplify by extracting perfect-power factors: $\sqrt{72} = \sqrt{(36 \times 2)} = 6\sqrt{2}$.

Scientific notation

Write a number as A × 10■, where 1 ≤ |A| < 10. Thus 0.00056 = 5.6 × 10■■ and 7,200,000 = 7.2 × 10■.

Worked example: Simplify $x^3 \times x^■ \div x^2$. Add exponents for multiplication and subtract for division: $x^{3■■■2} = x^■$. Also, $27^{2/3} = (\sqrt[■]{27})^2 = 3^2 = 9$.

8. Polynomials

Definition and degree

A polynomial is a finite sum of terms whose variable exponents are non-negative integers. Its degree is the largest exponent with a non-zero coefficient. A degree-1 polynomial is linear; degree 2 is quadratic; degree 3 is cubic.

Adding and subtracting

Arrange terms by powers and combine like terms. Subtracting a polynomial requires changing the sign of every term in the subtracted polynomial.

Multiplication

Use distribution and collect like terms. A polynomial multiplied by a monomial requires multiplying each term.

Remainder theorem

When a polynomial f(x) is divided by (x - a), the remainder is f(a). This can save time compared with long division.

Worked example: Add $(3x^2 + 2x - 1)$ and $(x^2 - 5x + 4)$: $4x^2 - 3x + 3$. For $f(x) = x^2 + 3x + 5$, the remainder on division by $x - 2$ is $f(2) = 4 + 6 + 5 = 15$.

9. Quadratic Equations

Standard form

A quadratic equation is $ax^2 + bx + c = 0$, where $a \neq 0$.

Factorisation

Factor the quadratic into two linear factors and use the zero-product rule: if $AB = 0$, then $A = 0$ or $B = 0$.

Quadratic formula

For $ax^2 + bx + c = 0$, $x = [-b \pm \sqrt{(b^2 - 4ac)}]/(2a)$.

Discriminant

$\Delta = b^2 - 4ac$. If $\Delta > 0$ there are two distinct real roots; if $\Delta = 0$ there is one repeated real root; if $\Delta < 0$ there are two non-real complex conjugate roots.

Completing the square

Rewrite $x^2 + bx + c$ as $(x + b/2)^2 + c - b^2/4$. This helps solve equations and identify the vertex of a quadratic graph.

Worked example: Solve $2x^2 + 3x - 2 = 0$. Here $a = 2$, $b = 3$, $c = -2$. $\Delta = 9 + 16 = 25$. $x = (-3 \pm 5)/4$, so $x = 1/2$ or $x = -2$.

10. Inequalities and Absolute Value

Inequality rules

Solve much like an equation. Reverse the inequality sign when multiplying or dividing both sides by a negative number.

Intervals

$x > 2$ is $(2, \infty)$; $x \geq 2$ is $[2, \infty)$; and $-1 < x \leq 4$ is $(-1, 4]$. Round brackets exclude endpoints; square brackets include them.

Absolute value

$|x|$ is the distance from zero and is never negative. For $c \geq 0$, $|x| = c$ gives $x = c$ or $x = -c$.

Worked example: Solve $-3x + 6 > 12$. Subtract 6: $-3x > 6$. Divide by -3 and reverse the sign: $x < -2$. Solve $|x - 3| = 5$: $x - 3 = 5$ or -5, giving $x = 8$ or $x = -2$.

11. Functions and Graphs

Function notation

A function assigns exactly one output to each permitted input. If $f(x) = 2x + 3$, then $f(4) = 11$.

Domain and range

The domain is the set of allowed inputs; the range is the set of outputs. For $f(x) = 1/(x - 2)$, $x = 2$ is excluded from the domain.

Linear graphs

A linear function has form $y = mx + c$. m is the gradient and c is the y-intercept. The gradient through two points is $(y■ - y■)/(x■ - x■)$.

Quadratic graphs

$y = ax^2 + bx + c$ is a parabola. It opens upward if $a > 0$ and downward if $a < 0$. Its axis of symmetry is $x = -b/(2a)$.

Composite and inverse functions

For $f(g(x))$, apply g first, then f. An inverse reverses a one-to-one function's input-output relationship.

Worked example: For $f(x) = 3x - 2$, $f(5) = 13$. The gradient through (1, 3) and (5, 11) is $(11 - 3)/(5 - 1) = 2$.

12. Rational Expressions and Equations

Simplifying

Factorise numerator and denominator, then cancel common factors. Values that make the original denominator zero remain excluded even if a factor cancels.

Solving

Find restrictions first. Multiply every term by the common denominator, solve the resulting equation and check that the solution is allowed.

Important caution

Do not cancel terms across addition or subtraction. For example, $(x + 2)/x$ cannot be reduced by cancelling x from only one term in the numerator.

Worked example: Simplify $(x^2 - 9)/(x^2 + 5x + 6)$. Factor: $[(x - 3)(x + 3)]/[(x + 2)(x + 3)] = (x - 3)/(x + 2)$, with $x \neq -3$ and $x \neq -2$.

13. Algebraic Modelling

A modelling process

Identify the unknown and define a variable. Translate the information into an equation or inequality. Solve, check the result against the situation and state units.

Percentages

Percentage change = (change/original) × 100%. After an increase of r%, multiply the original amount by $1 + r/100$. After a decrease, multiply by $1 - r/100$.

Rates and costs

A fixed fee plus a cost per unit can be modelled as total cost = fixed fee + (rate × number of units).

Worked example: A taxi charges ■500 plus ■150 per kilometre. For d kilometres, $C = 500 + 150d$. For 8 km, $C = 500 + 1{,}200 =$ ■1,700. A rectangle with width w and length $w + 3$ has perimeter 30: $2w + 2(w + 3) = 30$, so $w = 6$ cm and length = 9 cm.

14. Common Errors and Study Strategy

Frequent mistakes

Watch negative signs; distribute across every term; combine only like terms; reverse inequalities after division by a negative; and never divide by zero.

Check your work

Substitute a proposed solution into the original equation. Check whether denominators are non-zero and whether the answer makes sense in the word problem.

Study effectively

Practise actively. Close the guide and reproduce a rule from memory, solve a new problem, explain your method aloud and revisit mistakes after a day or two.

Practice Assessment

Attempt these questions before consulting the answer key. Show your working and state restrictions where relevant.

1. In $9x^2 - 4x + 12$, identify the coefficient of x and the constant.

2. Simplify $8a - 3 + 5a + 11$.

3. Evaluate $3x^2 - 2y$ when $x = -2$ and $y = 4$.

4. Expand $5(2x - 3)$.

5. Expand $(x + 3)(x - 4)$.

6. Factorise $12x^2 + 18x$.

7. Factorise $x^2 - 49$.

8. Solve $4x + 9 = 29$.

9. Solve $3(2x - 1) = x + 9$.

10. Solve $x + y = 9$ and $x - y = 1$.

11. Simplify $m■ \times m^3 \div m^2$.

12. Evaluate $64^{2/3}$.

13. Find the degree of $5x■ - 2x^2 + x - 8$.

14. Find the remainder when $x^2 - 2x + 4$ is divided by $x - 3$.

15. Solve $x^2 - 7x + 12 = 0$.

16. Solve $2x^2 - 5x - 3 = 0$.

17. Solve $-2x + 4 \geq 10$.

18. Solve $|x + 2| = 6$.

19. If $f(x) = 3x - 2$, find $f(5)$.

20. Find the gradient through (2, 4) and (6, 12).

21. Simplify $(x^2 - 16)/(x + 4)$, stating the restriction.

22. A number plus three times itself is 52. Find the number.

23. An item costs ■12,000 and is discounted by 15%. Find the sale price.

24. A rectangle has width w and length w + 5. Its perimeter is 34 cm. Find its dimensions.

Answer Key with Brief Working

1. Coefficient = -4; constant = 12.

2. 13a + 8.

3. $3(-2)^2 - 2(4) = 12 - 8 = 4$.

4. 10x - 15.

5. $x^2 - x - 12$.

6. $6x(2x + 3)$.

7. $(x - 7)(x + 7)$.

8. x = 5.

9. 6x - 3 = x + 9; 5x = 12; x = 12/5.

10. Add equations: 2x = 10, x = 5; then y = 4.

11. m■.

12. (■64)² = 4² = 16.

13. Degree 4.

14. f(3) = 9 - 6 + 4 = 7.

15. (x - 3)(x - 4) = 0; x = 3 or 4.

16. (2x + 1)(x - 3) = 0; x = -1/2 or 3.

17. -2x ≥ 6; reverse the sign: x ≤ -3.

18. x + 2 = 6 or -6; x = 4 or -8.

19. f(5) = 15 - 2 = 13.

20. (12 - 4)/(6 - 2) = 2.

21. (x - 4), with x ≠ -4.

22. n + 3n = 52; n = 13.

23. Discount = ■1,800; sale price = ■10,200.

24. 2w + 2(w + 5) = 34; w = 6 cm; length = 11 cm.

Formula Summary

Topic	Formula / rule
Distributive law	a(b + c) = ab + ac
Difference of squares	a² - b² = (a - b)(a + b)
Perfect square	(a ± b)² = a² ± 2ab + b²
Gradient	m = (y■ - y■)/(x■ - x■)
Quadratic formula	x = [-b ± √(b² - 4ac)]/(2a)
Discriminant	Δ = b² - 4ac
Index laws	a■a■ = a■■■; a■/a■ = a■■■
Percentage change	(change/original) × 100%

About this resource

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